Analytics Vidhya

Policy Evaluation and Policy Iteration

Instructor

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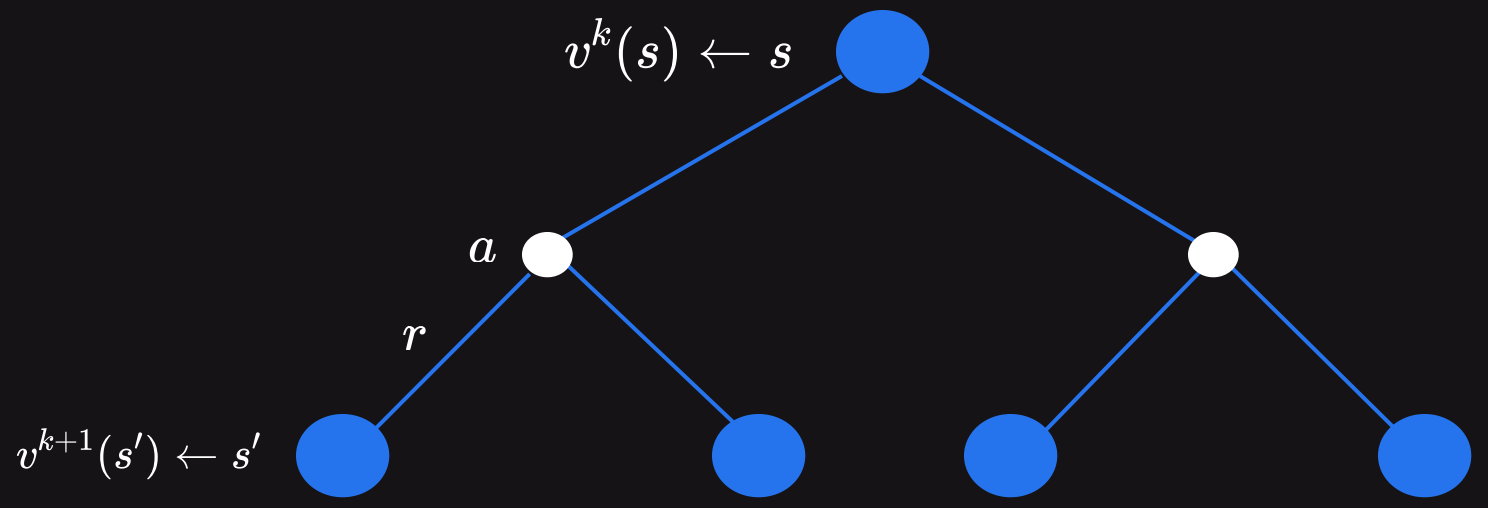
Policy Evaluation

$$v_\pi(s) = \sum_{a \in A} \pi(a|s) \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_\pi(s') \right)$$

Inspired from Bellman Expectation Equation



$$v^{k+1}(s) = \sum_{a \in A} \pi(a|s) \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v^k(s') \right)$$

$$v^{k+1} = R^\pi + \gamma P^\pi v^k$$

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Policy Evaluation

$$v^{k+1}(s) = \sum_{a \in A} \pi(a|s) \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v^k(s') \right)$$

Step 0

$$\text{Initialize } v_\pi^0(s) \leftarrow 0.$$

Step 1

$$v_\pi^1(s) \leftarrow \sum_{a \in A} \pi(a|s) \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_\pi^0(s') \right).$$

Bellman update/backup

Step k

$$v_\pi^{k+1}(s) \leftarrow \sum_{a \in A} \pi(a|s) \left(R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_\pi^k(s') \right).$$

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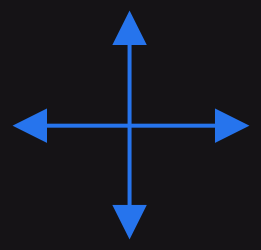
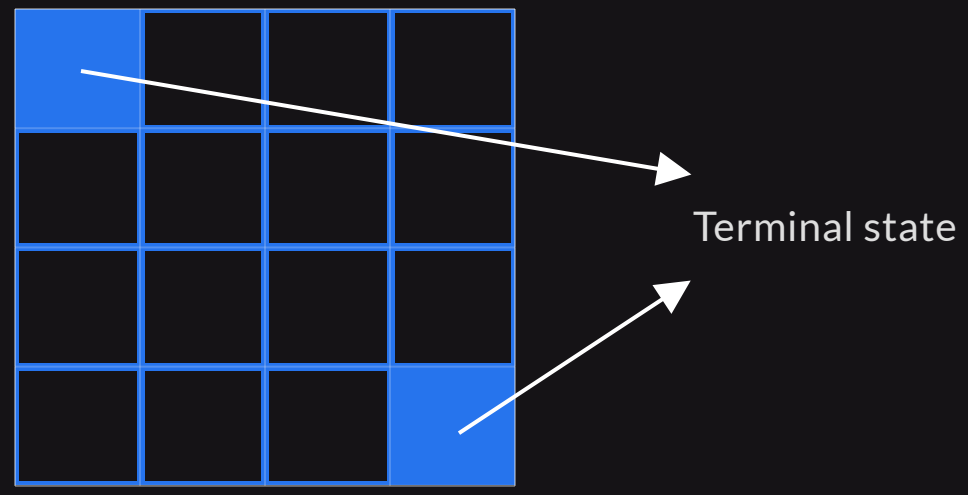
Policy Evaluation (Small Gridworld Example)



actions

$$r = -1$$

reward for every transition



Let us evaluate the uniform random policy. Assume y=1

$$\pi(\uparrow|\cdot) = \pi(\downarrow|\cdot) = \pi(\leftarrow|\cdot) = \pi(\rightarrow|\cdot) = 0.25$$

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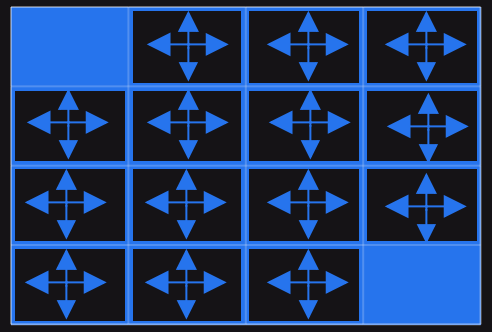
Small Gridworld Example

Returns

Policy

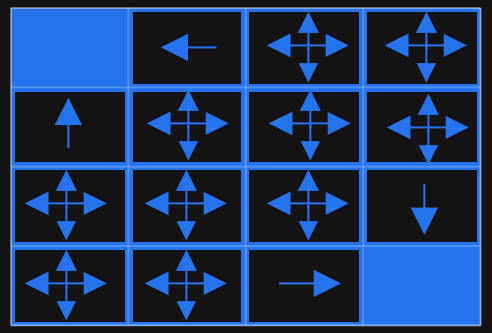
At $k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0



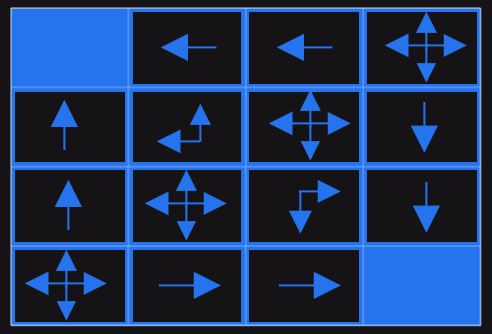
At $k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0



At $k = 2$

0.0	-1.7	-2.0	-2.0
-1.7	-2.0	-2.0	-2.0
-2.0	-2.0	-2.0	-1.7
-2.0	-2.0	-1.7	0.0



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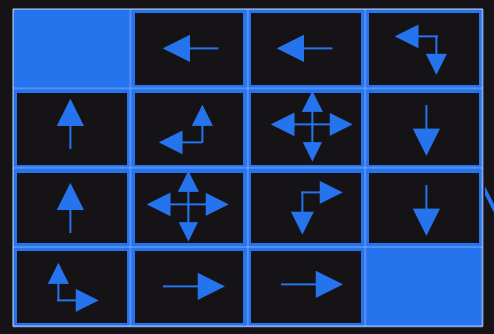
Small Gridworld Example

Returns

Policy

At $k = 3$

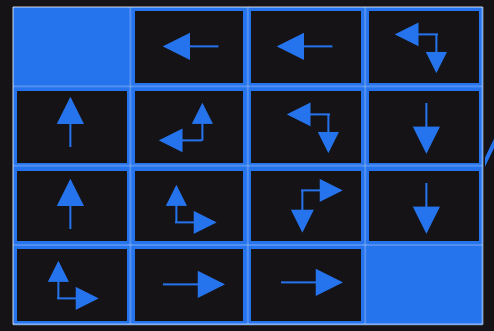
0.0	-2.4	-2.9	-3.0
-2.4	-2.9	-3.0	-2.9
-2.9	-3.0	-2.9	-2.4
-3.0	-2.9	-2.4	0.0



Optimal Policy

At $k = \infty$

0.0	-14.	-20.	-22.
-14.	-18.	-20.	-20.
-20.	-20.	-18.	-14.
-22.	-20.	-14.	0.0



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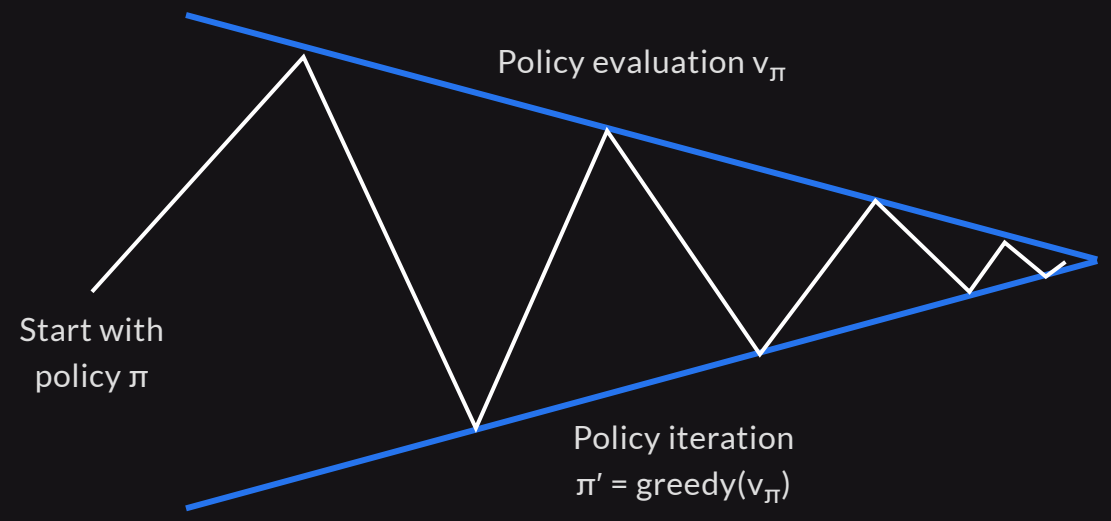
Policy Iteration (How to improve policy)

- Start with a policy π
  - Policy evaluation: Evaluate π using iterative policy evaluation
  - Policy improvement: Improve policy by acting greedily

$$\pi' = \text{greedy}(v_\pi)$$

- Set $\pi \leftarrow \pi'$
- Repeat above steps until convergence



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Thank You